

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester I)

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Class :FYIT	Batch : 2
Date of Assignment :21-07-2026	Date/Time of Submission :21/07/26

Scala For DS-7

Practical 7

Aim:Write a program to perform element-wise addition,subtraction,multiplication and division

```
import breeze.linalg._
object Pract7 extends App {
  val A = DenseMatrix((1.0,2.0,3.0),(4.0,5.0,6.0))
  val B = DenseMatrix((6.0,5.0,4.0),(3.0,2.0,1.0))

  val add = A + B
  val sub = A - B
  val mul = A *: B
  val div = DenseMatrix.tabulate(A.rows, A.cols){ (i, j)=> A(i,j) /
B(i,j)}

  println("Matrix A:")
  println(A)
  println("\nMatrix B:")
  println(B)

  println("\n Addition : ")
  println(add)
  println("\n Subtraction : ")
  println(sub)
  println("\n Multiplication : ")
  println(mul)
  println("\n Division : ")
  println(div)
}
```

Output:

Matrix A:

1.0 2.0 3.0
4.0 5.0 6.0

Matrix B:

6.0 5.0 4.0
3.0 2.0 1.0

Addition :

7.0 7.0 7.0
7.0 7.0 7.0

Subtraction :

-5.0 -3.0 -1.0
1.0 3.0 5.0

Multiplication :

6.0 10.0 12.0
12.0 10.0 6.0

Division :

0.16666666666666666 0.4 0.75
1.3333333333333333 2.5 6.0

Himanshu S038

Matrix A:

1.0	2.0	3.0
4.0	5.0	6.0

Matrix B:

6.0	5.0	4.0
3.0	2.0	1.0

Addition :

7.0	7.0	7.0
7.0	7.0	7.0

Subtraction :

-5.0	-3.0	-1.0
1.0	3.0	5.0

Multiplication :

6.0	10.0	12.0
12.0	10.0	6.0

Division :

0.16666666666666666	0.4	0.75
1.3333333333333333	2.5	6.0